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A Review on Predicting Failure in Industrial Machines: its Methods, Challenges and Future Direction

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***Abstract*— Predicting various system failures and optimizing maintenance strategies are important for improving reliability, safety, and efficiency, as well as reducing downtime of different machines such as metro trains, electric motors, industrial machinery, etc. This paper reviews 15 studies on failure classification and RUL prediction using various machine learning and deep learning algorithms and their combinations. The reviews have multiple limitations like imbalanced classes, large multivariate time series data, Feature selection, reliance on synthetic data, outlier detection, limited labeled data, missing data, etc. This paper primarily tries to recommend how to solve these issues and focuses on the advancements in predictive accuracy through the use of hybrid and ensemble models, as well as the significance of effective data collection and feature extraction. Also, the paper discusses the role of explainable AI methods in improving model interpretability which helps us in feature selection telling us which feature is contributing more to the failure. Future research directions aim to address these gaps by contributing to the development of robust, scalable, and interpretable failure prediction frameworks. However, to resolve various limitations identified, some can be addressed through the integration of hybrid models and explainable AI (XAI) methods, which this paper seeks to explore and recommend.**

Keywords—Failure classification, RUL prediction, Maintenance Optimization, Machine Learning, Deep Learning, Ensemble Models, Feature Extraction, Explainable AI (X-AI).

I. INTRODUCTION

The prediction of failures in industrial machines is crucial for maintaining operational efficiency, decreasing downtime, and improving safety in various sectors such as railways, manufacturing, and renewable energy systems. The integration of advanced digital technologies has enabled Industry 4. 0 to leverage artificial intelligence and machine learning for predicting failures and optimizing maintenance. Accurately forecasting failures, especially the remaining useful life (RUL) of machinery, allows industries to shift from reactive to proactive maintenance approaches, thereby reducing unexpected outages and expenses. This research investigates different techniques utilized for failure classification and RUL forecasting, including deep learning and hybrid ensemble models, which have shown significant potential across various industrial settings [1]–[6].

Recent improvements in failure forecasting have centered on employing time-series data analysis, hybrid ML-DL frameworks, and explainable AI methodologies. For instance, CNN-LSTM-based techniques have been effectively applied to estimate RUL in critical systems like aircraft engines [4] and industrial machinery [5]. Likewise, autoencoder-based anomaly detection has been employed for fault diagnosis in railway systems [1], while interpretable AI frameworks such as SHAP and LIME have tackled the requirement for stakeholder confidence in ML-generated predictions [7], [15]. Furthermore, innovative ensemble learning strategies have bolstered prediction robustness, particularly in instances involving imbalanced datasets, as demonstrated in maintaining photovoltaic systems [14] and semiconductor fabrication [8]. Nonetheless, challenges remain, including data imbalance, scalability, and the practical applicability of models, as discussed in research concentrating on diverse areas such as Scania truck systems [11], elevators [6], and software dependability [12].

The integration of explainable AI (XAI) has become a key area of research aimed at enhancing the interpretability of opaque models in predicting industrial failures. Methods like SHAP, LIME, and integrated gradient techniques offer insights into the significance of features and the decision-making processes of models [7], [15]. This is particularly essential for sectors that demand not just accurate forecasting but also actionable insights for making decisions. Moreover, initiatives to incorporate real-time data streaming and adaptive algorithms have created new opportunities for scaling predictive maintenance solutions in large, intricate systems [2], [5]. The potential for enhancing precision, effectiveness, and dependability through hybrid models and advanced preprocessing strategies further emphasizes the vital role of AI in revolutionizing maintenance approaches [9].

Despite these progressive developments, numerous research gaps continue to exist. A considerable number of current studies depend on synthetic or restricted datasets, limiting the generalizability of the models. Additionally, issues such as outlier detection, feature selection, and the scalability of hybrid models to extensive industrial systems have not been sufficiently tackled [10]. This paper recognizes these gaps and suggests novel solutions by combining hybrid ensemble models with explainable AI methods to improve both predictive accuracy and interpretability. In particular, we examine how merging XAI techniques like SHAP with sophisticated ML-DL architectures can bridge the divide between prediction effectiveness and model clarity. The research also highlights the significance of robust data collection, preprocessing methods, and adaptive learning to create scalable and industry-ready failure prediction systems. Through this thorough review, we aspire to provide actionable recommendations for overcoming existing limitations and

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establishing a roadmap for future research trajectories in the domain of industrial failure prediction.

II. Literature Review

N. Davari, B. Veloso, R. P. Ribeiro, P. M. Pereira, and J. Gama [1] discuss the focus on detecting failures and errors before they occur and doing maintenance for air production units in the railway industry to prevent unexpected breakdowns and minimize downtime. The study proposes a method that includes data-driven predicting failure and doing its maintenance framework based on anomaly detection using the DL method. It uses a sparse autoencoder network to detect abnormal data from analog and digital sensors on the air production unit system. Also, a low-pass filter is used for detecting the failure. One limitation includes the availability of limited labeled data for training, the use of real-time implementation, and managing a large volume of multivariate time-series data. Future directions include a focus on expanding the availability of labeled data, optimizing real-time implementation, and developing accurate methods for handling and visualizing large multivariate time-series data.

O. N. NEAMAH and R. BAYIR [2] discuss predicting faults in systems like MetroPT and metro PT-3 dataset in industry 4.0 using ML algorithms to improve the reliability of the system and its maintenance process as well as detecting faults precisely before they occur. The method used in the study is a framework of ML algorithms with state-of-the-art classification, including RL, KNN, SVM, etc. These algorithms are used for prediction with confusion matrix reports and metrics like accuracy, precision, etc. One limitation of the study is not exploring the scalability of the proposed framework, which shows high accuracy in the dataset and its real-world applications. To overcome limitations future direction can be focusing on testing the framework on large datasets or complex models in different industries. Also, explore the integration of real-time streaming data and complex models in the framework.

A. Shinde, M. Motiramani, P. Soni, and A. Nanade [3] discuss predicting the failure and processing its maintenance of the rail transportation industry using ML and DL algorithms which can be user-friendly and cost-effective to reduce the wastage of valuable time and resources in the industry. The method includes algorithms like LSTM, random forest, variational autoencoders, and boost. This study also focuses on the application of time-based features using lag and rolling statistics to analyze time series data. Limitations of the study include as many industries do not have a complete system for predicting when machines will fail which leads to wastage of time and resources during unexpected failures, the trade-off between precision and recall in the models, as well as the impact of sample size and standard deviation on the results. Future directions include creating a smart approach like the metaheuristic method to overcome the study limitations and apply it to real train hubs in different locations for practical testing, researchers could consider adding factors like risk and uncertainty into the process of how to maintain train tracks and explore using advanced techniques like deep learning.

A. P. Hermawan, D. -S. Kim, and J. -M. Lee [4] discusses predicting the remaining useful life of aircraft engines and also focuses on providing real-time information about engine conditions due to the low tolerance of engines where minor faults can lead to huge accidents. The method includes a combination of CNN and LSTM models which learn the behavior of historical data and provide accurate information about the time to failure of the system. CNN layer helps the system learn the features of historical data while the LSTM layer can memorize past information for predicting future data. Limitations of the study include a focus on single engine data which does not fully reflect real-world conditions and also the comparison conducted in previous studies is mostly based on the modification of their algorithms but they did not consider any more complex data as used in the current study. Future directions can include exploring more real-world data and also exploring energy-efficient approaches that can be used in the future.

J. Du, X. Gui, L. Song, S. Gao, C. Zhang, and X. Li [5] discuss predicting failure and its maintenance in mechanical equipment for more accurate prediction of remaining useful life to reduce downtime and costs related to failures. The method proposed in the study is the Transferable CNN-LSTM model which is an integration of CNN and LSTM networks for feature extraction from time-series sensor data. Domain-adversarial training of Neural Networks (DANN) is used for model portability in different equipment conditions. One limitation of the study include not validating in real industrial settings and does not focus on issues related to data quality and practical implementation challenges. Future directions include a focus on validation using real-world datasets, including domain knowledge into the model, developing advanced data preprocessing techniques, and exploring adaptive methods or approaches to continuously learn and adapt to changing industrial conditions for predicting future failure in the equipment.

G. Verma, J. Awatramani, and N. Hasteer [6] discuss predicting failure and doing its maintenance in elevator systems which will focus on the importance of safety and reliability in modern buildings and the need for effective maintenance to minimize costs. The methods include using DL techniques like LSTM, using sensor data and a deep autoencoder model for feature extraction, and training a random forest classifier for predicting faults in the elevator. One limitation includes possible uncertainties and unforeseen conditions which will impact the reliability of the model. It also does not focus on the influence of holidays on elevator usage which will affect maintenance planning. Future research includes solving uncertainties in the model and considering including holiday data to improve maintenance planning accuracy. Also, exploring adaptive approaches to address dynamic operating conditions which will improve the model's reliability.

S. Gawde, S. Patil, S. Kumar, P. Kamat, K. Kotecha, and S. Alfarhood [7] discuss model prediction of industrial rotating machines data as we can predict future failure in the machine and it is a challenge to understand in a human-friendly way the AI/ML model internal decision So to overcome this challenge they have used X-AI models. This study also involves Multi-sensor-based multi-fault data acquisition, Frequency-domain statistical feature extraction as well Comparison of results from multiple AI models, and a combination of X-AI techniques like LIME, SHAP with PDP, and ICE for interpretation of the models. This paper lacks the use of a combination of multiple X-AI models to improve the internal decision of models as well as machine data containing only singular fault types rather than focusing on multiple fault types. The future direction for this paper based on its limitations can be exploring the advantages of combining

multiple X-AI techniques, focusing on the interpretation of predictions for multiple fault types, and developing more complex and high-performance models and their interpretations.

D. Pradeep, B. V. Vardhan, S. Raiak, I. Muniraj, K. Elumalai, and S. Chinnadurai [8] explore the difficulties encountered by semiconductor manufacturing firms in optimizing the identification and forecasting methods for manufacturing failures and their maintenance to enhance production without compromising quality or raising production expenses. The primary issue of this study is the elevated downtime rate of wafer fabrication equipment, with a significant portion linked to unplanned maintenance. This leads to decreased efficiency and higher costs for both the manufacturers and the consumers. In this study, various ML and neural network algorithms are employed for predicting manufacturing failures and their maintenance. They utilized the secom dataset from the UCI repository, which comprises 1567 records of semiconductor manufacturing, and also implemented processes such as data cleaning, attribute selection, model development, and model evaluation. The key restriction of this study is its concentration on utilizing algorithms to minimize unexpected maintenance in semiconductor manufacturing, neglecting the main factors contributing to resource shortages and health management issues that result in production delays. Furthermore, the paper does not address the potential challenges or constraints of applying the suggested maintenance algorithms or techniques in actual production settings. In light of this limitation, future research could analyze the causes of limited resources and inadequate health management affecting the company's production, along with the practical implementation of ML algorithms or hybrid methodologies for failure prediction in semiconductor manufacturing to achieve improved accuracy and efficiency.

M. N. Al-Andoli et al. [9] propose an ensemble model combining hybrid machine learning (ML) and deep learning (DL) algorithms to improve fault prediction in industrial machines. The model uses a parallel ensemble approach with two base models and one meta-model, processed in a parallel computation environment. The base models incorporate hybrid backpropagation and particle swarm optimization to optimize features and parameters. Evaluated on a standard dataset, the model demonstrated improved prediction accuracy and reduced processing time. The research addresses the labor-intensive nature of previous failure prediction methods by leveraging AI algorithms. However, the reliance on gradient descent in DL may lead to high computational costs and suboptimal solutions. Future work could explore more efficient optimization techniques and additional AI algorithms for enhanced accuracy.

T. Son Pham, D. Huy Nguyen, and N. Duy Minh Phan [10] discuss the classification of failure and problems related to its maintenance in ball bearing systems which will increase the availability of the system, downtime reduction, and reduce the cost of maintenance. This study uses interpretable ML techniques to have high accuracy in classifying failure and also understanding which features are influencing the model results. They used method of ensemble methods like Random Forest, Gradient Boosting, and AdaBoost for their prediction and also used interpretable algorithms like shap in the model to explain which is the most important feature for which type of failure using local and global analysis for the dataset of the system. One possible limitation of the study is that it uses random forest techniques which have high computational time as it uses and computes multiple trees which can be difficult to interpret using the interpretation model mentioned before. Also, the study needs a flexible algorithm that can adapt to the changes in system operation over time. Based on limitations future direction of the paper can be developing a flexible algorithm that can adapt to the changes in the system operation over time and also focus on improving interpretable algorithms that are applied in the model for classifying failure using high dimensional data.

K. T. Selvi, N. Praveena, K. Pratheksha, S. Ragunanthan, and R. Thamilselvan [11] discuss the problem of predicting failure and their maintenance for air pressure systems mainly air brake systems in Scania trucks. The data collected from the real world can have imbalanced data where each classes are not uniformly distributed which can lead to errors in detecting or classifying failure in the systems. To resolve this issue and balance the classes, resampling techniques like over- and under-sampling and SMOTE are used. The study uses classification algorithms like Logistic Regression, Random Forest, Naïve Bayes, SVM, etc. for calculating accuracy in predicting failures in the systems. One limitation can be to improve the model performance study can use attribute selection techniques to select the most important attributes for prediction and also problems in handling imbalance real-world data before model prediction. The future direction of the study can be using an advanced attribute selection method to find out which attribute is contributing more to the failure in the brake systems.

E. Elahi, S. Kanwal, and A. N. Asif [12] discuss improving the prediction of failure in software using the ensemble method as software datasets contain imbalanced data which will improve the quality of software and reduce the costs of its maintenance. The study improves the performance of the prediction model by selecting the optimized combination of classifiers and resampling techniques in the model averaging method. The study uses ensemble methods like model averaging, stacking, and voting to improve the performance of the prediction model. The study uses a combination of resampling techniques like smote, and ros with different classifiers like logistic regression, KNN, and decision tree for making final predictions. Statistical tests like the Friedman test and ANOM test are used to determine the importance of results and determine the best combination of classifier and resampling technique. One limitation of the study is not using word software datasets for general results as the experiment uses datasets from the PROMISE repository. Also, some classifier is selected which can limit the experiment for other combinations to improve prediction accuracy. Based on the limitations future direction of the study could be using real word software datasets for general results, developing a more effective ensemble method of combination of classifiers and resampling techniques for failure prediction, and research predicting different types of faults using different ensemble methods.

S. Bundasak and P. Wittayasirikul [13] discuss the development of predicting failure in motors, their health, and their maintenance using AI. The study predicts damage and lifespan of motors that convert electricity into mechanical energy for the production process. The study detects malfunction of the motor early before it fails and notifies the maintainers through a website which will reduce the motor

downtime and it will be more effective in the maintenance process. The study includes a method of analyzing the data and its electrical parameters collected from the operating motor. The study develops the website system using Python, PHP, and Java scripts and a logistic regression model and compares the performance of different algorithms like K-nn, Logistic Regression, Naïve Bayes, and SVM for motor health prediction. One limitation of the study is the difficulty in collecting additional test data, and the training data collection process is time-consuming. The study also mentioned that if test data is not in the training data it cannot be predicted and the system will not perform well using large datasets or large noise datasets. Future directions include collecting more datasets and training data via APIs to reduce time consumption. Also study suggests that the collection of test data should not require a lot of effort or resources if the collection of data is easier the conduction of the test will be more effective.

R. B. Mofidul, S. S. Alam, A. Chakma, B. Chung, and Y. M. Jang [14] talk about how to predict failures and maintenance in solar energy systems by using group algorithms to estimate maintenance schedules with a small number of features found in the systems. The research employs a 99.9 kW solar panel system dataset and concentrates on variables such as DC voltage, DC, instantaneous power output, power factor, and frequency. The research primarily aids in creating a system that determines the most effective group algorithm through performance measurements and identifies key features like DC voltage and DC for model forecasting. The approach taken in the research involves creating a data frame from the solar system data, choosing significant features, and dividing the dataset into training and testing data at a 4:1 ratio. Each group model such as RandomForest, XGBoost, CatBoost, and LightGBM is utilized for training and subsequently for testing the data for predictions, with XGBoost showing the best results, achieving an accuracy of 98.62% and a precision of 94.37%. Ultimately, performance indicators like accuracy and precision are applied to evaluate the outcomes of each model. The research gap identified in the study is the application of ensemble techniques using a minimal set of features for maintenance forecasting in these systems. Considering limitations, upcoming directions may involve research and enhancements in ensemble techniques for predictive modeling in solar energy systems, as well as future inquiries into utilizing extra features and alternative algorithms within the research framework for improved predictions and maintenance.

T. Khan, K. Ahmad, J. Khan, I. Khan, and N. Ahmad [15] talk about the difficulty of predicting failures in industrial work by emphasizing the prediction of the time until failure, known as Remaining Useful Life (RUL) of equipment. Since industries largely rely on advanced machinery, keeping track of and ensuring good machine performance over time becomes a very crucial job. Standard machine learning methods are useful in terms of performance metrics such as Mean Squared Error (MSE), but they operate as a "black box," which is not very clear for the stakeholders' confidence in their predictions. To tackle this problem, the authors suggest a clearer regression framework that incorporates X-AI methods mainly LIME and SHAP. This framework aims to enhance understanding and trust by offering insights into the prediction process. The approach consists of two primary elements: selecting features and making predictions based on machine learning, along with the clarification of these predictions utilizing X-AI models. LIME provides local clarity by assessing how model predictions change with alterations in input examples, while SHAP delivers both local and global interpretations that provide a complete view of the model's functioning. The article also points out some limitations, such as the balance between the precision of machine learning models and their clarity. The complexity of certain models might make it hard to achieve both outstanding performance and understandable explanations. Furthermore, the research mainly concentrates on LIME and SHAP without considering other X-AI methods, and the conclusions may be constrained by the particular datasets utilized. Future studies could fill these gaps by looking into more XAI techniques like Grad-CAM or DiCE, creating algorithms that find a balance between accuracy and transparency, testing the framework in various industrial settings, and customizing explanations to fit the requirements of different stakeholders, thus improving decision-making processes.

III. Analysis and Discussion

The progress of failure prediction, driven by machine learning and deep learning, has considerably changed how industries observe and handle equipment failures. By utilizing time-series sensor data, operational records, and sophisticated machine-learning algorithms, organizations can foresee failures before they happen, which leads to considerable cuts in downtime, enhanced safety, and better resource distribution. This section examines the techniques and results of different studies in this field and discusses the suggested approach for creating strong predictive models. The foundation of any failure prediction system relies on the information collected from machinery. In this research, data were collected from multiple sources, including time-series sensor data, maintenance logs, and operational records, to form a well-rounded dataset for training the model. The sensor data, which usually encompasses readings from both analog and digital sensors, observes variables like temperature, pressure, vibration, and RPM of machinery. Maintenance records are essential for understanding failure trends and predicting when maintenance work is needed. Research, such as those by Davari et al. [1] and Du et al. [5], highlights the significance of gathering real-time sensor data for precise predictions. Nevertheless, ensuring the quality of data can be difficult, particularly in sectors where sensor failures or data transfer problems may lead to incomplete or incorrect data.

Data preprocessing is a crucial phase in constructing failure prediction models. Raw sensor data frequently contains missing values, noise, duplicate values, and outliers, which can negatively influence the effectiveness of models. In this study, various preprocessing methods were used to tackle these problems, following recommendations from Shinde et al. [3] and Al-Andoli et al. [9]. Missing values were handled using interpolation or backfilling to maintain continuity in the time-series data. Outliers were detected and eliminated using statistical approaches like the z-score and Interquartile Range (IQR), while normalization methods such as Min-Max scaling and Z-score normalization were utilized to standardize the features to a common scale, aiding in better convergence in both ML and DL algorithms. These preprocessing actions ensured that the data was clean, standardized, and well-suited for training the model. Feature engineering and selection were the subsequent essential steps in model development. New features were generated from raw sensor data to uncover hidden patterns in system behavior. Time-related features,

such as rolling means, moving averages, and lag features, were developed to capture temporal dependencies in the data, as mentioned by Shinde et al. [3] and Hermawan et al. [4].

After relevant features were generated, feature selection methods such as Recursive Feature Elimination (RFE), mutual information, and feature importance from tree-based techniques (e.g., Random Forest) were implemented to lessen the dimensionality of the dataset and enhance model performance, by techniques used by Gawde et al. [7].

In the area of developing models, a mix of traditional machine learning (ML) techniques, deep learning (DL) architectures, and hybrid approaches were applied. The research started with ML techniques like Random Forest, gradient boosting, bagging, and stacking to set a baseline for performance evaluation. These techniques are easy to understand and can act as useful benchmarks. More sophisticated deep learning architectures, which include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), Gated Recurrent Units (GRUs), and Temporal Convolutional Networks (TCNs), were used for their capability to model intricate temporal relationships found in time-series data. LSTMs, notably, have shown significant success in predicting failures in systems such as aircraft engines [4] and machinery [5]. Moreover, hybrid models, like mixtures of CNN and LSTM networks, were created to utilize CNN's feature extraction abilities and the LSTM's skill in capturing long-term dependencies in sequential data. From this examination, a hybrid model incorporating TCN and GRU can be developed to address any limitations. These hybrid models showed better performance, especially in noisy or high-dimensional data settings, as demonstrated in research by Du et al. [5].

Evaluating models is crucial to determine the trustworthiness and accuracy of predictions. In this research, models were assessed using different metrics, such as accuracy, precision, recall, F1-score, and mean squared error (MSE), following the methods outlined by Shinde et al. [3] and Khan et al. [15]. Accuracy and precision measure the model's general prediction performance and its ability to correctly identify failures, while recall and F1-score offer insights into the model's ability to prevent false negatives—vital in predictive maintenance to reduce downtime. For ongoing prediction tasks, such as forecasting Remaining Useful Life (RUL), MSE, RMSE, and MAE were employed. To enhance model performance, hyperparameter tuning was carried out using Grid Search and Random Search, as recommended by Hermawan et al. [4] and Al-Andoli et al. [9]. These techniques optimized the model parameters, ensuring peak performance.

Given the importance of model interpretability in industrial applications, Explainable AI (XAI) techniques were incorporated into the study. SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-agnostic Explanations), and integrated gradients were used to provide transparency into the models' decision-making processes, as demonstrated by Khan et al. [15] and Verma et al. [6]. SHAP and LIME were particularly valuable in explaining the contribution of individual features to the predictions, which is crucial in environments where stakeholders need to understand the rationale behind maintenance recommendations. Integrated Gradients were applied to neural networks to trace the influence of each feature, improving trust in the predictions made by complex deep learning models. These XAI techniques not only helped increase stakeholder trust but also provided valuable insights into system behavior, thereby informing better maintenance strategies and decision-making.

After model development and evaluation, the models were deployed for real-time monitoring in the operational environment. Real-time sensor data was continuously fed into the models, and alerts were triggered whenever a failure was predicted. These alerts enabled proactive maintenance scheduling, an advantage highlighted by Gawde et al. [7] and Selvi et al. [11]. Automated alert systems were developed to notify maintenance teams of potential failures, allowing them to take preventive actions and avoid costly downtime. To ensure that the models remained accurate over time, a continuous retraining framework was implemented. As more data became available, the models were updated to adapt to new patterns, ensuring that the predictive maintenance system remained effective and relevant to changing conditions.

IV. Conclusion

This review has examined the significant role of machine learning (ML) and deep learning (DL) techniques in predicting failures, estimating Remaining Useful Life (RUL), and optimizing maintenance schedules within industrial systems. The methodologies discussed, including time-series data collection, advanced preprocessing, feature engineering, and the application of ML models, have shown promise in improving system reliability and efficiency. However, several gaps and challenges remain that must be addressed to further enhance the efficacy of these predictive models.

One major gap identified is the need for more adaptable algorithms that can evolve with changing system operations. Current models, such as Random Forest and ensemble methods, are effective in handling imbalanced classes and large multivariate time-series data but are computationally expensive and difficult to interpret, particularly with high-dimensional data. Future research should focus on developing more flexible algorithms that can be updated in real-time, ensuring continued predictive accuracy as system behaviors change. Additionally, improving interpretability is crucial for gaining stakeholder trust in model predictions.

Another challenge is the difficulty in handling imbalanced and noisy real-world datasets. While techniques like SMOTE and data resampling are used to address these issues, synthetic data generated through these methods may not always be reliable for failure prediction. There is a need for more robust data preprocessing techniques, including time-series resampling to reduce data volume, and feature selection approaches, such as LIME and SHAP, to identify the most influential features after model implementation. Advancements in noise reduction and data quality improvements will help mitigate this issue, particularly in real-time settings.

Given the dynamic nature of industrial systems, continuous model retraining is essential to maintain accuracy as data evolves. Developing frameworks for real-time monitoring and automatic retraining will ensure models remain effective over time. Additionally, incorporating diverse datasets across multiple industries will help develop more generalizable models, making these techniques applicable to a wider range of use cases.

Finally, future research can be considerable potential in combining traditional ML methods with advanced DL architectures, such as Temporal Convolutional Networks (TCNs) and Gated Recurrent Units (GRUs), to address some of the existing gaps. Integrating explainable AI (XAI) techniques, like LIME and SHAP, into hybrid models can improve feature extraction and provide insights into which features contribute most to failure predictions. This combination of ML and DL approaches, along with XAI, could lead to more accurate, interpretable, and adaptable models.

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